Informatique / iut de BORDEAUX

M2201 - Language Theory

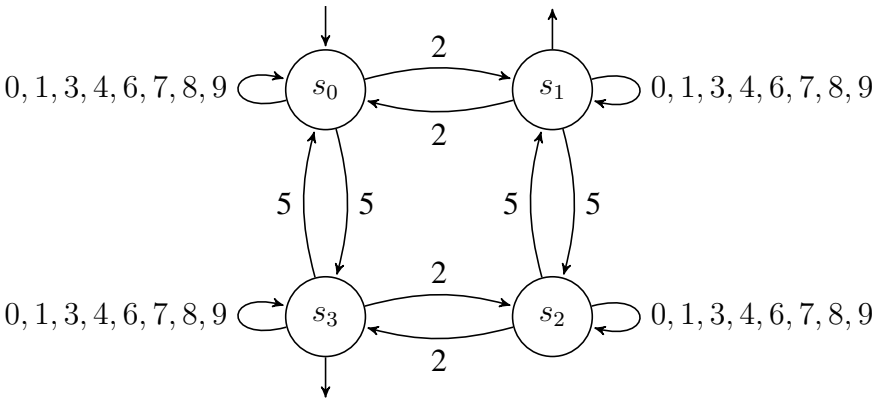
Introduction

TD1

« Machines » recognizing numbers...

The following directed graph describes a machine that works as follows:

- s_0 is the start vertex (denoted by an incoming arc from "outside").
- The machine reads an input word (given separately). It reads it one character at a time, and each time, we move to a neighbor vertex according to the corresponding arc.
- When the entire word has been read, if the current vertex is a terminal vertex (denoted by an outgoing arc to the "outside", here s_1 or s_3), then the machine answers "YES" (we say that it *recognizes* or *accepts* the word), otherwise it answers "NO" (we say that it *rejects* it).



For example, if the above machine reads the word 3563572, its behavior is the following:

$$s_0 \xrightarrow{3} s_0 \xrightarrow{5} s_3 \xrightarrow{6} s_3 \xrightarrow{3} s_3 \xrightarrow{5} s_0 \xrightarrow{7} s_0 \xrightarrow{2} s_1$$

Since the last vertex s_1 is a terminal vertex, the word 3563572 is accepted.

1 What does this machine do?

- Try to guess what words are accepted by the above machine (describe them with a sentence)
- Is this machine as simple as it could be?

2 Now, let's build some of these machines...

Try to build a machine that recognizes the following words:

- Words that represent even numbers
- Words that represent multiples of 10
- Words that represent multiples of 100 (there is a subtlety)

- Words that contain two consecutive identical digits, such as 422, 16623445, or 43331, but neither 1231 nor 6789. You don't have to represent all the states explicitly (dots $\ldots$ are allowed), but your picture should be clear.
- Words that contain 153 as a substring, such as 312153 and 915321, but neither 98762 nor 13513
- Strings that do not contain 153 as a substring
- Strings that represent multiples of 3